



PEM 2025 Mathematics Advanced Trial HSC Examination

Mathematics Advanced HSC (Australian Islamic College of Sydney)



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Student Number

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2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning
- and/or calculations

Total marks: 100**Section I – 10 marks (pages 2-7)**

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8-32)

- Attempt Questions 11-35
- Allow about 2 hours and 45 minutes for this section

Directions to School or College

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Section I

10 marks

Attempt Questions 1-10

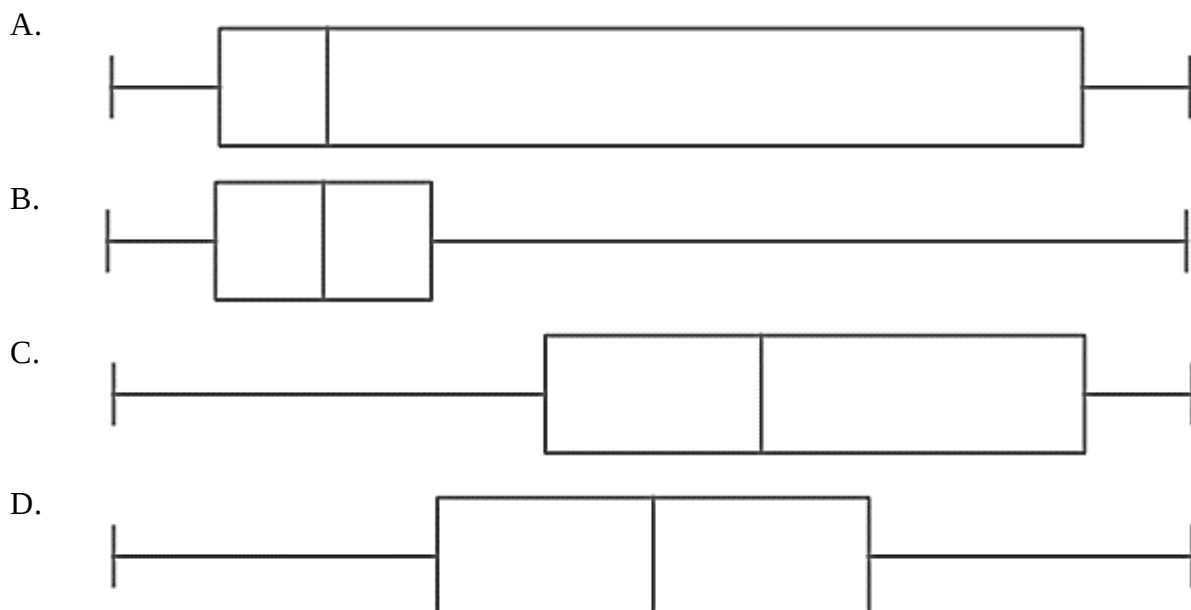
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. The stem-and-leaf plot below shows the distribution of the hourly wages (\$) of a manufacturing company.

4	0 0 2 2 2 4 4 5 7
5	0 0 1 2 3 6 8 9 9
6	3 6 6 8
7	1
8	3 7
9	6

Which if the following box-and-whisker plots may represent the distribution of their hourly wages?



2. Two fair dice are thrown in a game. If the sum of the two numbers thrown is 7, the player gains \$36. The player gains \$6 on any other number rolled.

What is the expected gain from this game?

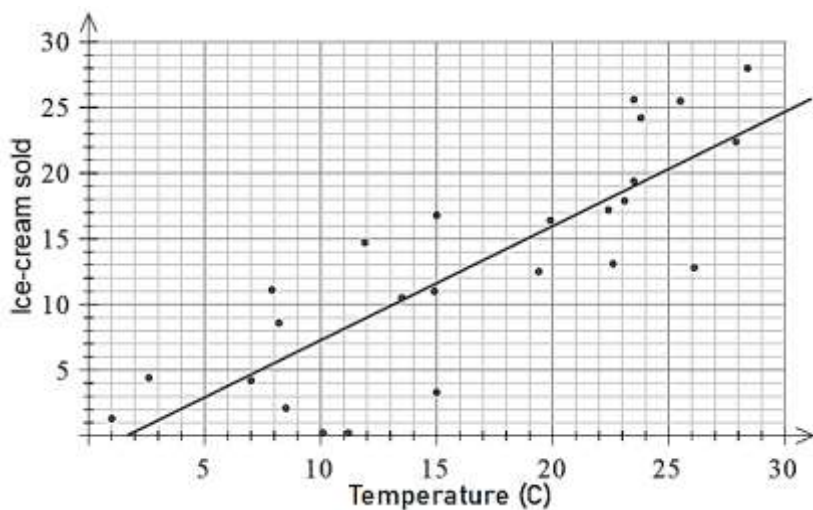
- A. \$11
- B. \$12
- C. \$30
- D. \$31

3. If $\log_{10}(x-a)=3$, what is the value of x ?

- A. a^3
- B. $1000a$
- C. $1000+a$
- D. $30+a$

4. The scatterplot below displays the average number of ice-creams sold against the daily temperature (C°).

A least squares regression line has been fitted to the scatterplot.



What is the equation of the least squares regression line closest to?

- A. ice cream = $-1.333 + 0.867 \times \text{temperature}$
- B. ice cream = $-7.077 + 1.154 \times \text{temperature}$
- C. temperature = $-1.333 + 0.867 \times \text{ice cream}$
- D. temperature = $-0.867 + 1.333 \times \text{ice cream}$

5. Let $0 < q < p$. If p and q are the 1st and 2nd terms of a geometric progression. What is the limiting sum?

- A. $\frac{p}{1-q}$
- B. $\frac{pq}{q-p}$
- C. $\frac{p^2}{p+q}$
- D. $\frac{p^2}{p-q}$

6. Given that $\sin \theta = \frac{2}{5}$ and $0^\circ < \theta < 90^\circ$.

What is the value of $\sin(90^\circ - \theta) + \sin(180^\circ + \theta)$?

- A. $\frac{-\sqrt{21}-2}{5}$
B. $\frac{2-\sqrt{21}}{5}$
C. $\frac{\sqrt{21}-2}{5}$
D. $\frac{\sqrt{21}+2}{5}$

7. Given that $\int_2^8 f(x) dx = C$ and $\int_6^8 f(x) dx = D$.

What is the value of $\int_2^6 f(x) dx$?

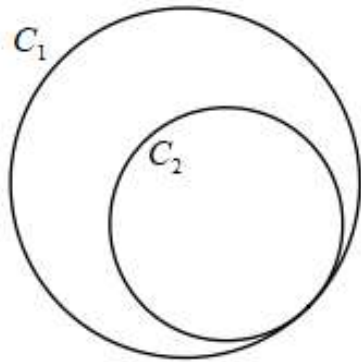
- A. $C+D$
B. $C+D-2$
C. $C-D$
D. $D-C$

8. The table below shows the centres and radii of two circles C_1 and C_2

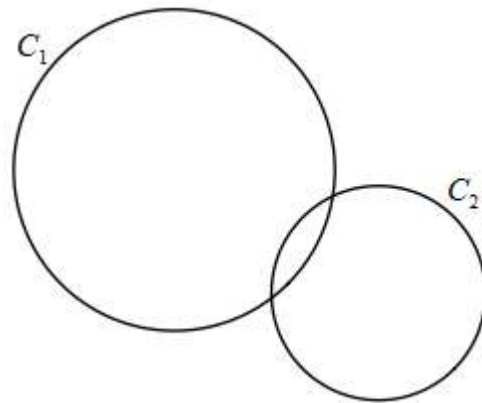
	Centre	Radius
C_1	$(2, 2)$	3
C_2	$(5, -2)$	2

Which of the following may represent the graphs of both C_1 and C_2 ?

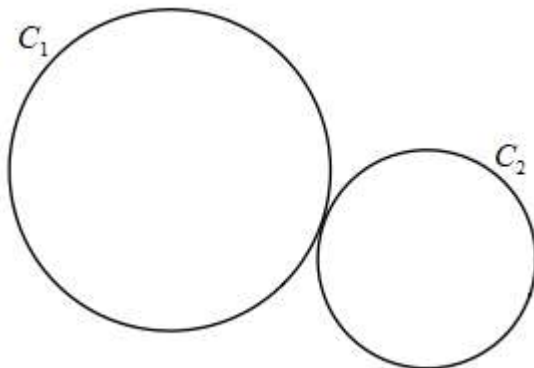
A.



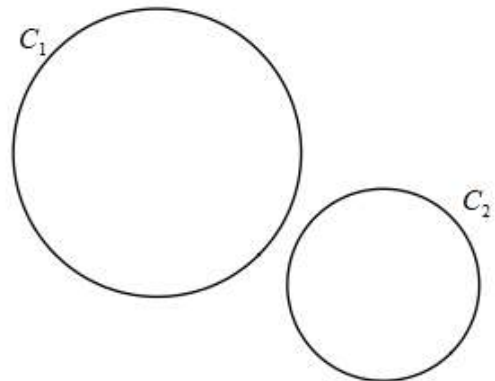
B.



C.



D.



9. If $f(x) = \tan(g(x^2))$, where $g(x)$ is a differentiable function, what is $f'(x)$?

A. $2x \sec^2(g(x^2))$

B. $2xg'(x^2) \sec^2(g(2x))$

C. $2xg(x^2) \sec^2(g(x^2))$

D. $2xg'(x^2) \sec^2(g(x^2))$

10. Let $f(x) = \ln\left(x + \frac{\sqrt{3}}{2}\right)$, $x \in (-\infty, \infty)$ and $g(x) = \cos x$, $x \in (-\infty, \pi)$.

What is the largest domain where both $f(g(x))$ and $g(f(x))$ exist?

A. $\left(-\frac{\sqrt{3}}{2}, \frac{5\pi}{6}\right)$

B. $\left(-\frac{\sqrt{3}}{2}, \frac{2\pi}{3}\right)$

C. $\left(-\frac{5\pi}{6}, \frac{5\pi}{6}\right)$

D. $\left(-\frac{5\pi}{6}, -\frac{\sqrt{3}}{2}\right)$

Trial Higher School Certificate Examination

Mathematics Advanced

Section II Answer Book

90 marks

Attempt Questions 11-35

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.
- Your response should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (3 marks)

The n th term of a sequence is given by $T_n = 7n - 3$.

- (a) Write down the first 3 terms of the sequence. 1

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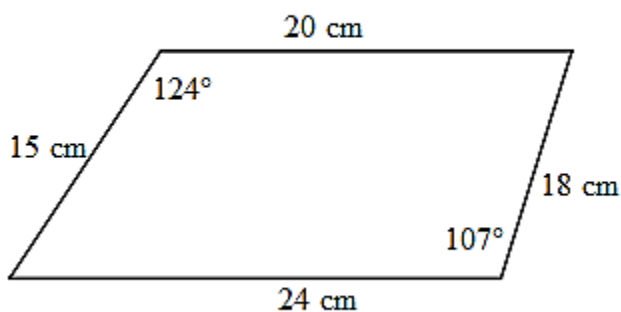
- (b) Find the sum of the first 100 terms of the sequence. 2

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Question 12 (2 marks)

Find the area of the shape below to 2 decimal places. 2

NOT TO SCALE



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Question 13 (4 marks)

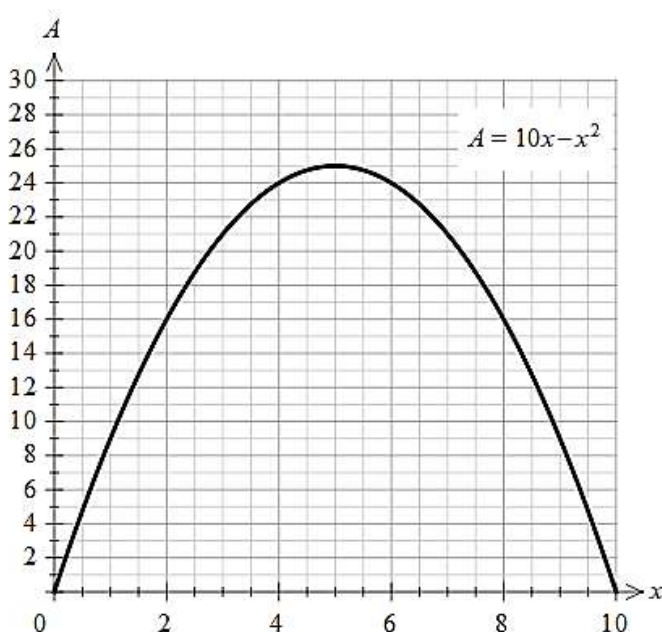
A piece of wire 20 cm long is bent into a rectangle.

Let one side of the rectangle be x cm long.

- (a) Show that the area, A can be expressed by $A = 10x - x^2$. 1

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The graph of $A = 10x - x^2$ is shown below.



- (b) Using the graph, or otherwise, determine the value(s) of x when the area is 16 cm^2 . 1

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- (c) Find the dimensions of the rectangle when the area is at its maximum. 2

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Question 14 (3 marks)

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Two families are purchasing tickets for a theme park online.

The first family purchased 2 adult and 3 student tickets for a total of \$190.

If one adult ticket costs \$ x and one student ticket costs \$ y . Then the following equation can be formed for the first family:

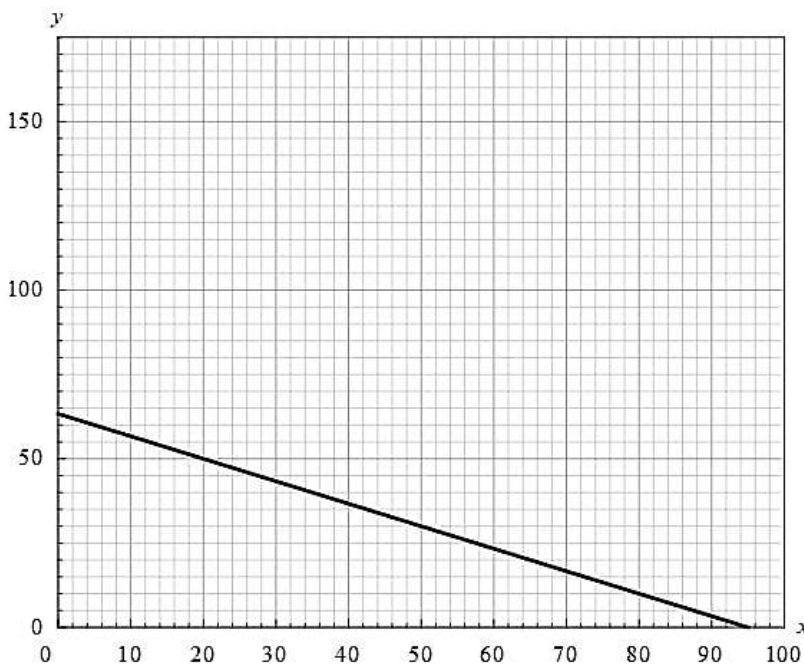
$$2x + 3y = 190$$

The second family purchased 5 adult and 2 student tickets for a total of \$310.

The graph for the first family is sketched below.

By writing another relevant equation for the second family and graphing it on the same grid find the cost of one adult and one student ticket.

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Cost of one adult ticket = _____

Cost of one student ticket = _____

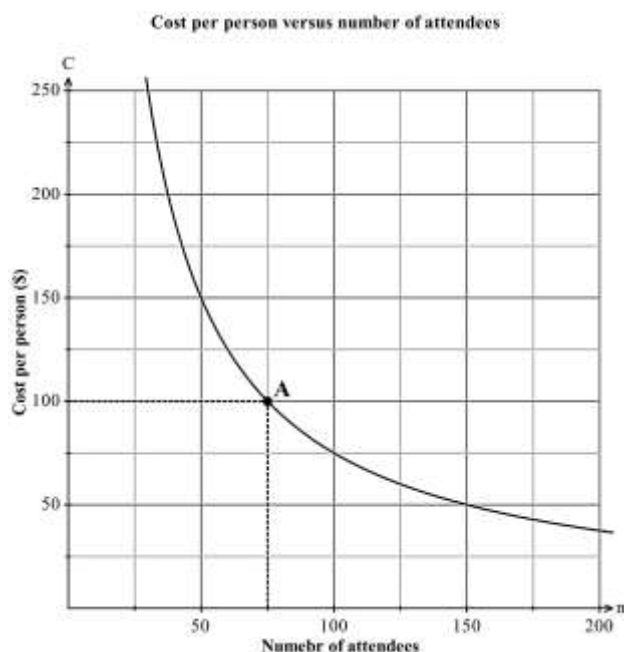
Question 15 (4 marks)

The cost of hiring a chartered bus for a return trip from Sydney to Brisbane is \$7500.

The cost will be shared equally by the people attending the trip, so that C (\$) is the

cost per person when n people attend the trip. The graph below shows the relationship

between the cost per person and the number of people attending the trip.



The graph is a hyperbola because C is inversely proportional to n .

- (a) The point A lies on the hyperbola. Find the equation relating C and n . 2

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- (b) Give one limitation of this equation relation to this context. 1

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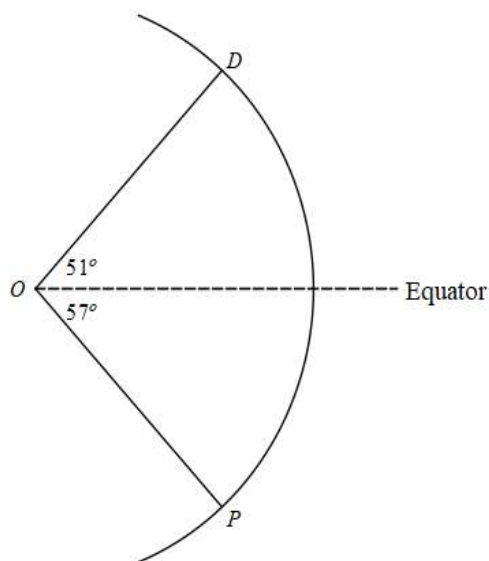
- (c) It is possible for the cost per person to be \$80? 1

Support your answer with appropriate calculations.

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Question 16 (3 marks)

D'Alembert and Poincare are two craters located on the Moon. Both craters lie on the same line of longitude. As shown in the diagram D'Alembert Crater (D) lies 51° North of the Moon's equator, while Poincare Crater (P) lies 57° South of the Moon's equator. The centre of the Moon is O .



- (a) What is the size of $\angle DOP$ in radians? 1

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- (b) If the radius of the Moon is 1740 km, find the surface distance between the two craters to the nearest kilometre. 2

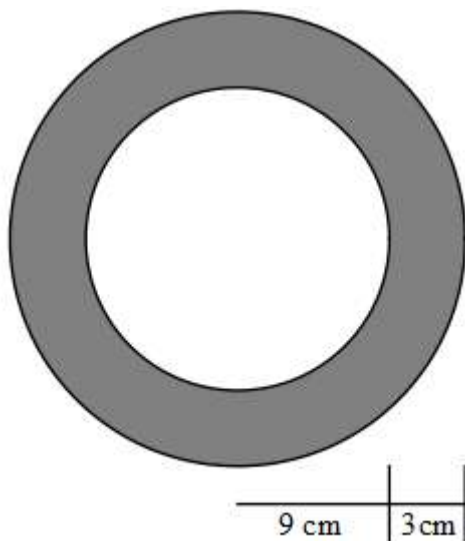
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Question 17 (3 marks)

The figure below shows a circular dartboard with two concentric circles.

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The radii of the circles are 12 cm and 9 cm respectively.



Two darts are thrown and hit the dartboard.

Is the probability higher if both darts land in the shaded region, or if only one dart lands in the shaded region?

Justify your answer with relevant calculations.

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Question 18 (2 marks)

Find the equation of the tangent to the curve $y = x^3 - 3x$ at $x = 1$. 2

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Question 19 (2 marks)

Find $\int \frac{4x}{e^{x^2}} dx$. 2

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Question 20 (2 marks)

Show that $\sec^2 A \equiv \frac{\operatorname{cosec} A}{\operatorname{cosec} A - \sin A}$. 2

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Question 21 (4 marks)

Tony takes out a reducing balance loan of \$240 000, with an interest rate of 4.8% p.a. compounded monthly.

Tony makes regular monthly repayments.

The following table shows the first 3 months

Payment Number	Payment Paid (\$)	Interest Paid (\$)	Principal Reduction (\$)	Balance (\$)
0	0.00	0.00	0.00	240 000.00
1	2741.05	960	1781.05	238 218.95
2	2741.05	952.88	1788.17	236 420.78
3		A		B

- (a) Determine the values of A and B. 2

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- (b) If Tony repays the loan in 9 years, calculate how much interest Tony pays in total over the life of the loan. 2

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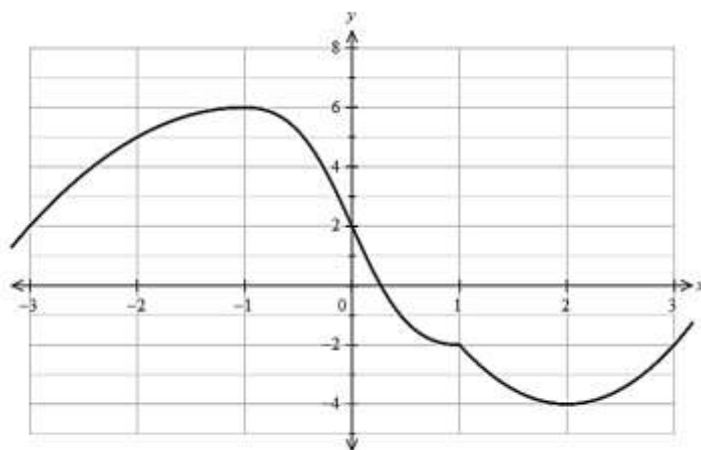
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Question 22 (2 marks)

The graph of $f(x)$ is shown below.

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Approximate $\int_{-2}^2 f(x)$ using four rectangles of equal length and the right endpoint of each rectangle.

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Question 23 (2 marks)

Differentiate $y = \ln\left(\frac{x^2}{\sqrt{x+1}}\right)$, leaving your answer in most simplest form.

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Question 24 (3 marks)

- (a) Show that $\frac{x+7}{x-1}$ can be expressed as $1 + \frac{8}{x-1}$. **1**

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- (b) Hence, or otherwise, evaluate $\int_2^e \frac{x+7}{x-1}$, leave your answer in exact form. **2**

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Question 25 (2 marks)

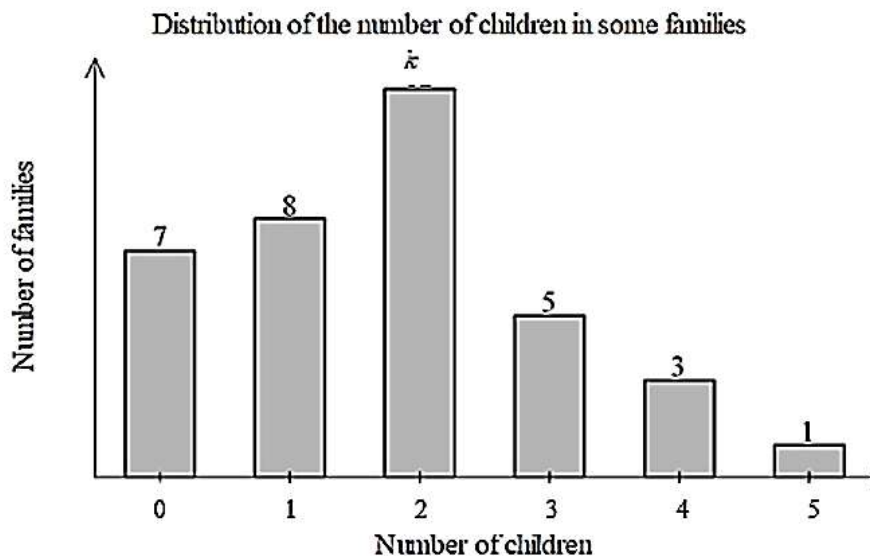
Sketch a graph of a continuous function with the following properties, **2**

- $f(x)$ is odd
- $f(3) = 0$ and $f'(1) = 0$
- $f'(x) > 0$ for $x > 1$
- $f'(x) < 0$ for $0 \leq x < 1$



Question 26 (4 marks)

The bar chart below shows the distribution of the number of children in some families of a building.



If a family is randomly selected from the families, the probability that the selected family has less than 3 children is $\frac{3}{4}$.

- (a) Show that $k = 12$. 1

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- (b) How many families are within one standard deviation of the mean number of children? 3

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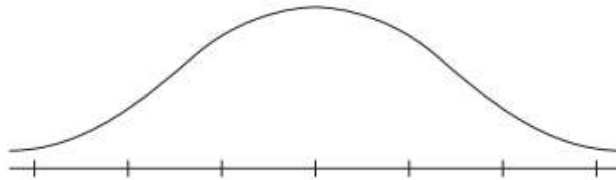
Question 27 (4 marks)

The mean and standard deviation of the test marks of a class are μ and σ respectively.

The table below shows the test marks and the standard z-scores of 2 students, Chloe and Tiffany.

	Test mark	Z-score
Chloe	42	-0.5
Tiffany	60	1

- (a) By labelling the normal distribution curve below, or otherwise, find the values of μ and σ . 2



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- (b) The result of a student who got 48 was not included in the calculations. 2
How will the z-scores of Chloe and Tiffany change when the missing mark is now included?

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Question 28 (4 marks)

- (a) Sketch the curve $g(x) = 2 \sin\left(\frac{\pi x}{2}\right)$ from $x = -2$ to $x = 2$. **2**

- (b) Evaluate the exact area bounded by the curve $g(x)$ and the x axis. **2**

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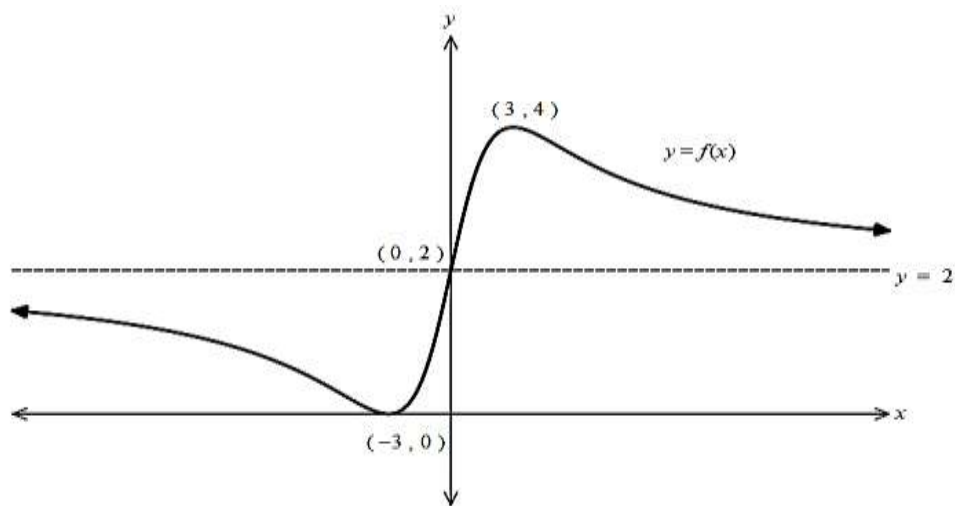
Question 29 (3 marks)

The figure below shows the graph of a curve with equation $y = f(x)$.

The curve meets the x axis at $(-3, 0)$ and the y axis at $(0, 2)$.

The curve has a maximum at $(3, 4)$ and a minimum at $(-3, 0)$.

The line $y = 2$ is a horizontal asymptote to the curve.



- (a) What is the range of $\frac{1}{2} f(x)$? 1

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- (b) Sketch the graph of $g(x) = f(3x) - 2$, labelling all points and asymptotes. 2

Question 30 (4 marks)

A particle moves along a straight line such that its distance x , in metres, from a fixed point O is given by $x = 1 - 2 \sin\left(2t + \frac{\pi}{6}\right)$, where t is measured in seconds from $t = 0$.

- (a) Where is the particle initially? **1**

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- (b) When and where, does the particle first come to rest? **2**

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- (c) What is the exact acceleration of the particle when $t = \frac{\pi}{12}$? **1**

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Question 31 (7 marks)

For the curve $y = x^4 - 2x^2$.

- (a) Find any stationary points and determine their nature. **3**

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- (b) Find any points of inflexion. **2**

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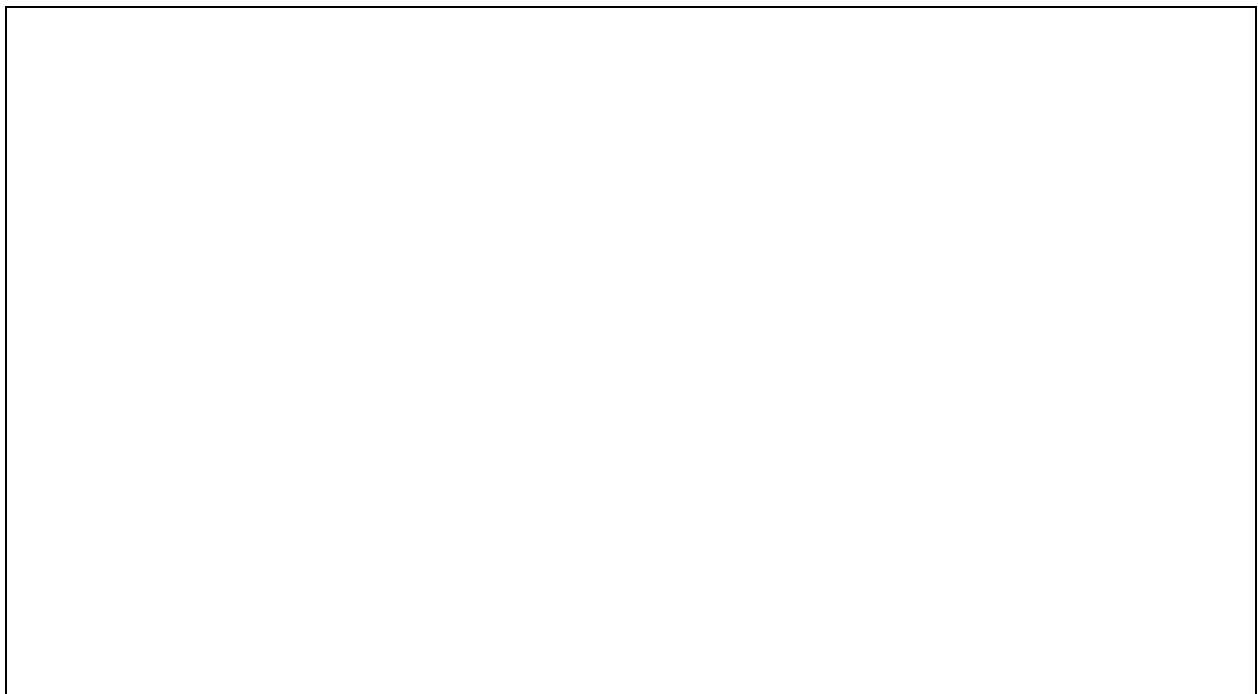
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Question 31 continues on page 25

Question 31 continued

- (c) Sketch the curve in the domain $-2 \leq x \leq 2$ showing all stationary and inflexion points, as well as any intercepts with the axes. **2**

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End of Question 31

Question 32 (4 marks)

A quadratic has equation $y = A + Bx - x^2$.

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When the function is reflected along the y axis, it is identical to a horizontal translation of -3 units in the x axis.

Given that the curve meets the y axis at $(0,18)$, determine the area of the region bounded by the curve and the x axis.

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Question 33 (5 marks)

A continuous random variable X has the following cumulative distribution function $F(x)$, defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ Bx^2(4 - x^2) & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

- (a) Show that $B = \frac{1}{3}$. **1**

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- (b) Find $P(X > 0.5)$. **1**

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Question 33 continues on page 28

Question 33 continued

A skewness coefficient can be calculated by the formula,

$$\frac{\text{mean} - \text{mode}}{\text{standard deviation}}.$$

- (c) You are given that $E(X) = \frac{28}{45}$ and $E(X^2) = \frac{4}{9}$. **(DO NOT PROVE THIS)** 3

Evaluate the skewness coefficient for this distribution and interpret your result.

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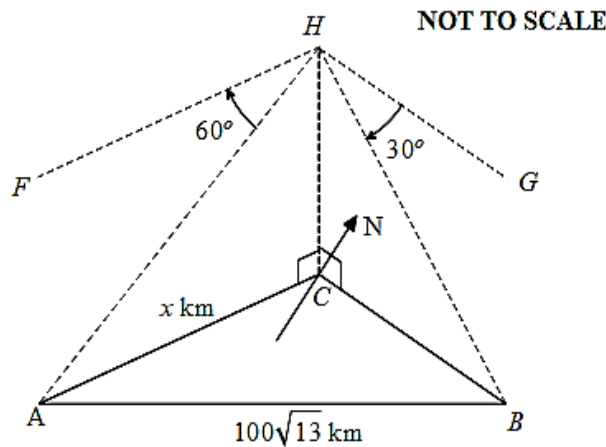
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End of Question 33

Question 34 (7 marks)

The figure below shows three buoys at sea at A , B and C . A helicopter is flying along FH and then HG , where F, G and H are three points of the same height vertically above A, B and C respectively. When the helicopter reaches H , the angles of depression of A and B from the helicopter are 60° and 30° respectively, while the true bearings of A and B from C are 250° and 130° respectively. The distance between A and B is $100\sqrt{13}$ km.



- (a) Show that the distance between A and C is 100 km. 3

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Question 34 continues on page 30

Question 34 continued

At 9 a.m., two destroyers start travelling from A and B . The destroyer at A travels along AC with a uniform speed of 20 km/h , while the destroyer at B travels along BC with a uniform speed of 50 km/h .

- (b) Show that the distance between the two destroyers after travelling for t hours is $10\sqrt{1300 - 450t + 39t^2}$. 2

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- (c) The destroyers are known to be “on alert” when they are less than 40 km apart. 2
During what times will the two destroyers be “on alert?”

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End of Question 34

Question 35 (7 marks)

Charlene wants to buy a new house and she joins a savings plan offered by a bank. The bank pays interest at a rate of 6% p.a., compounded monthly. On the first day of January 2025, Charlene places an initial deposit of \$5 000 into her bank account. On the first day of every month afterwards she deposits 1% more than she did in the previous month.

- (a) Show that the amount in Charlene's account at the end of the n th month after 3

1st January 2025 is $\$1\,005\,000\left((1.01)^n - (1.005)^n\right)$

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Question 34 continues on page 32

Question 34 continued

A house in Sydney is worth \$3 000 000 on 1st January 2023. The selling price of the house grows by 0.5% every month.

- (b) Find the selling price of the house at the end of the n th month since Charlene started her savings plan. **1**

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- (c) The deposit for buying the house is 30% of its selling price. At the end of which month and year will Charlene have enough money for a deposit of the house. **3**

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End of Exam

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- 33 -

Section II extra writing space

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